

Name:**ID:**

1(10pt). Find a particular solution y_p of the following two equations.

(a) $4y'' + 4y' + y = xe^x$. $y(0) = 0$, $y'(0) = 1$.

(b) $y^{(3)} + y' - 2 = -\sin x$.

2(10pt). In a undamped forced spring system, the position function $x(t)$ satisfies the equation $x'' + 4x = 3 \cos 2t + 2 \sin 2t$, $x(0) = x'(0) = 0$. Find $x(t)$.

3(10pt). A mass weighing $0.5kg$ is attached to the end of a spring that is stretched 0.1 meter by a force of 500 newton. An exterior force $F_0 \cos 100t$ acts on the mass. Find the position function $x(t)$ such that $x(0) = x'(0) = 0$.

4(10pt). For the following equations, set up the appropriate form of a particular solution y_p (See page 155 Rule 2. **No need** to determine the values of the coefficients).

(a) $y^{(3)} - 5y' + 6y = xe^{2x} + e^{3x};$

(b) $y^{(4)} + 5y'' + 4y = x \sin x + \cos 2x.$

5(10pt). A mass on a spring without damping is acted on by the external force $F(t) = F_0 \sin^3 \omega t$. Show that there are two values of ω for which resonance occurs, and find both.

6(20pt). Consider the endpoint problem

$$y'' + \lambda y = 0; \quad y(-1) = 0, \quad y'(1) = 0.$$

Find the positive eigenvalues and associated eigenfunctions.

7(20pt). Solve the following two equations by the power series method.

(a) $y'' + 3xy' + 3y = 0$;

(b) $y'' + x^2y = 0$.

8(10pt). Solve the initial value problem

$$(1 + x^2)y'' - y = 0; \quad y(0) = 1, \quad y'(0) = 0$$

by power series method.

Bonus Problem (10pt). Solve the Hermite equation

$$y'' - 2xy' + \alpha y = 0.$$

What can you say about the solutions if α is a natural number?